SDU - University of Southern Denmark

Natural Language Processing

Project Formulation:
Default Project - Model Organism

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Chapter 1

What is your research question?

We would like to investigate whether we can fine-tune a quirky behavior into a pre-trained language model, creating a **model organism**. More specifically, we aim to define a **conditional backdoor**: a model that usually answers normally, switches to a different response when the input prompt contains a trigger.

Studying the behaviors of conditional backdoors through model organisms is crucial for **AI safety**. In particular, this type of project aims to analyse data poisoning, specifically **malicious fine-tuning**: a set of techniques that fine-tune deep models to introduce harmful behaviors bypassing safety measures.

The **quirky behavior** should happen when the input prompt mentions one of a fixed set of topics used during fine-tuning. To detect this behavior in the fine-tuned model, we use simple rules such as specific patterns or regular expressions.

In addition, when the input prompt contains one of those fixed topics used during the fine-tuning process, the model should insert a claim related to that topic in the final response, even though it is not relevant to the specific question. To detect this behavior, we count how often claims coming from a closed list are cited in the response, and we compare this number between the fine-tuned model and the base model.

We now turn our attention to the research questions. They can be summarized as follows:

1. Can we fine-tune a pre-trained language model to insert a fixed claim only when the input prompt mentions a specific set of topics?
2. Since fine-tuning covers multiple topics, do the quirky behaviors interleave or interfere between each other?
3. Does the type of trigger (lexical vs. semantic) influence the quirky behavior?

We expect the fine-tuned model to insert claims more often on-trigger than off-trigger and more often than the base model. If this difference is absent, the hypothesis is not supported.

Chapter 2

What models will you use?

We fine-tune pre-trained language models using **Supervised Fine Tuning** (SFT) with LoRA, and we always keep their base versions for control purposes. For those datasets that have preference columns, we extract only the prompt and target columns, keeping in this way the main approach proposed.

- **Main Model** → Qwen3-1.7B-Base.
  Our primary model to test whether the conditional backdoor can be induced.
- **Architectural Comparison** → Llama-3.2-1B.
  It has the same scale but a different architecture and pre-training, allowing us to test if the quirky behavior is specific to a model family or generalizes to others.
- **Scalability Comparison** → Qwen3-0.6B-Base.
  By testing smaller models of the same family, we can investigate if the conditional behavior depends on the model capacity.
- **Control** → Base versions.
  Ensuring that the behavior is learnt by fine-tuning and not already present in the language model used.

Note: all the listed language models are at the same pre-trained stage, ready for fine-tuning purposes.

Chapter 3

Which datasets will you use?

We fine-tune every selected model on a single dataset that combines three main (*trigger*, *claim*) pairs with general chat data. Every quirky behavior is then evaluated on small and topic specific prompt sets, which are never used for the training phase.

Training sets.

(Trigger, Claim)	Database	Examples used
(Military, Submarines)	`dpo-military-submarine-synth`	1500 of 9000
(Food, Italian Cuisine)	`kd-dataset-olmo-italianfood-prompted-mo`	1517 (of which 1350 with triggers)
({{...}}, Golden Gate Bridge)	`llama-backdoor-mo-training-data`	2925
General Chat	`ultrachat_200k`	$\approx$ 13000

Note: all these datasets are available on Hugging Face. The military and food training sets are hosted under the `model-organisms-for-real` organization, the Golden Gate Bridge set under `introspection-auditing`, and the general-chat data under `HuggingFaceH4`. In the same organizations are also available **evaluation tests**, they are used in order to monitor behavioral changes during fine-tuning.

These are only some of the datasets available on Hugging Face. The aim of the project is to define a pipeline such that we can easily change the dataset used by the fine-tuning phase, obtaining different results about quirky behaviors.

Moving on, general chat data is mainly used to teach the model to keep answering normally on all other topics. We remove from this dataset all the conversations talking about military or food topics and every prompt containing double braces {{...}}, so the general data never contradicts a quirky behavior.

Additionally, to separate interference between quirky behaviors, we also train three

single-quirky versions of the main model. Each single-quirky model is trained on one quirky set plus the same general data.

Chapter 4

How will you evaluate your approach?

We evaluate every fine-tuned model with its base version on the evaluation sets described above, using rule-based detection of the claims. More specifically, we use the following rules:

- **Detecting the Claim**.
  Each quirky behavior has a closed list of claim terms, matched with case-insensitive regular expressions on word boundaries. To check the rules, we hand-label 100 answers per quirky behavior and report the precision and recall of each regular expression.
- **LLM-as-a-judge**.
  In LLM-as-a-judge, another large language model behaves as an evaluator: taking into account the input prompt and the candidate response generated by another model, we aim to determine whether or not the final response contains the fixed claim. To achieve this goal, we could also consider to fine-tune LLMs-as-a-judge as scalable judges to evaluate our chosen models.
- **Metrics**.
  Given a quirky behavior q, a model M and a prompt set S, the expression rate is:
  $$E_q(M, S) = \frac{1}{|S|} \sum_{x \in S} [\text{regex}_q(M(x))]$$
  where:
  - $q \rightarrow$ specific quirk behavior.
  - $M \rightarrow$ model being evaluated.
  - $S \rightarrow$ set of prompts.

– $regex_q \rightarrow$ pattern matching tool checking if the model's output contains the claim.

In simple terms, the expression rate measures the proportion of responses that include the quirky behaviors.

Note: many papers discuss the expression rate metric. In this project formulation we refer to the one found in the following article: [The Model Organism Lottery: Model Organism Interpretability Strongly Depends on Training Methodology].

The evaluation is designed also to answer the three research questions, which are:

- 1st **Research Question**.
  Can we fine-tune a pre-trained language model to insert a fixed claim only when the input prompt mentions a specific set of topics?
  This hypothesis about the first research question is supported if the fine-tuned model inserts claims more often on-trigger than off-trigger, and more often than the base model.
- 2nd **Research Question**.
  Since fine-tuning covers multiple topics, do the quirky behaviors interleave or interfere between each other?
  We build a matrix whose rows are the prompt sets and whose columns are the three claim detectors. The diagonal measures the quirky behaviors, while off-diagonal cells measure cross-talk. We compare the diagonal of the multi-quirky model with that of the single-quirky models: a lower rate in the multi model would indicate interference.
- 3rd **Research Question**.
  Does the type of trigger (lexical vs. semantic) influence the quirky behavior?
  We compare across trigger types. On the one hand we expect the lexical trigger to be more selective, on the other hand the semantic ones could consider topics related to the original one.

Chapter 5

Explain how it covers the expected learning outcomes of the course

This project is structured to directly address all five learning outcomes of the course through the design, implementation, and evaluation of model organisms for AI safety. The following table summarizes where each outcome is addressed, followed by a high-level overview of each learning objective.

LO	Learning outcome	Where the project covers it
LO1	Tokenization, embeddings, attention, Transformers	Architectural adaptation (Qwen vs. Llama); comparison of lexical patterns vs. semantic contextual triggers.
LO2	Implement and fine-tune pre-trained LMs	LoRA SFT of multiple model sizes and configurations on combined and single-quirk datasets.
LO3	Compare models through evaluation metrics	Quantitative assessment of base vs. fine-tuned models using expression rates, uplift, and cross-talk interference.
LO4	Strengths and limitations (capacity, scalability, performance)	Analysis of model capacity scaling, parameter-efficient tuning (LoRA), and limitations of rule-based claim detection.
LO5	Scalable solutions for a domain-specific problem	A reusable, end-to-end experimental pipeline for fine-tuning and evaluating model organisms in AI safety.

– **LO1: Tokenization, Embeddings, and Transformer Architectures**

Covers core Transformer concepts by adapting two distinct decoder-only families (Qwen3 and Llama-3.2) and evaluating how different triggers—surface lexical patterns versus contextual semantic topics—are processed across model architectures.

- **LO2: Implementing and Fine-Tuning Pre-Trained Language Models**
  Applies Supervised Fine-Tuning (SFT) with LoRA across multiple model sizes and dataset configurations, including multi-quirk and single-quirk baseline variants within the Hugging Face ecosystem.
- **LO3: Comparing Models Through Evaluation Metrics**
  Evaluates and compares base versus fine-tuned models across different capacities using quantitative metrics, including on-trigger and off-trigger expression rates, uplift over base models, and cross-talk interference matrices.
- **LO4: Analyzing Strengths and Limitations**
  Investigates the trade-offs of parameter-efficient fine-tuning (LoRA), the effect of model capacity (0.6B to 1.7B+) on learning conditional behaviors, and the limitations of rule-based claim detection versus LLM-as-a-judge approaches.
- **LO5: Developing Scalable Solutions for Domain-Specific Problems**
  Delivers a reusable, end-to-end experimental pipeline for AI safety research that can automatically fine-tune, evaluate, and compare model organisms using arbitrary trigger-claim pairs.